

Classification and Regression on Real-World Verification and Exam Data

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Abstract

We present two end-to-end machine learning pipelines on real-world datasets. For classification, we predict student academic performance category (Average, Good, Very Good, Excellent) from the CEE Student Entrance Examination dataset using k-Nearest Neighbors, Logistic Regression, and Gaussian Naive Bayes. For regression, we predict auction verification runtime from the Auction Verification dataset using kNN, Support Vector Regression, and Random Forest models. All models were compared via 10-fold cross-validation and tuned with `GridSearchCV`. Logistic Regression had the highest classification accuracy (54.4%), while Random Forest delivered the best regression performance ($R^2 = 0.991$). Source code is available at <https://github.com/zachshotwell/AI-Project-GroupSCR>.

1 Introduction

Predicting outcomes from structured data is a major part of supervised machine learning. This report evaluates two such tasks drawn from real-world datasets. First, can demographic and study-habit features reliably categorize a student’s performance on a university entrance exam? Second, given the process capacities and property attributes of an online auction, can we accurately estimate its formal verification runtime?

Both tasks present modelling challenges. The CEE dataset contains four imbalanced performance classes and a mix of nominal and ordinal categorical features, requiring careful encoding and feature selection. The Auction Verification dataset has a continuous target with a pronounced right-skewed distribution, motivating a log transform on the target variable.

For each task we benchmark three models, select the strongest baseline, tune all models with `GridSearchCV`, and evaluate the best result on a held-out test set. We report accuracy and macro-averaged F_1 for classification; R^2 and RMSE for regression.

2 Data

2.1 Dataset Description

CEE Student Performance. An ARFF-format dataset originally containing 695 student records, each described by demographic attributes (gender, caste) and academic features (coaching, class percentages). After removing 73 exact duplicate rows (10.5% of the original data), 622 clean records remained. The target variable **performance** has four ordered classes: Average, Good, Very Good, and Excellent. Classes are imbalanced; Excellent is the smallest group (15.8%).

Auction Verification. A CSV dataset with 2,043 instances and seven features: four process capacity measurements and three property attributes. The continuous target **verification.time** records auction verification runtime in seconds. No duplicate rows were found.

2.2 Exploratory Summary

Table 1 shows the class distribution of the CEE target variable. Good is the plurality class (31.2%); Excellent is the minority class (15.8%), giving a 2:1 imbalance between the largest and smallest classes.

Table 1: CEE target class distribution ($n = 622$ after deduplication).

Performance	Count	Proportion
Good	194	0.312
Vg	178	0.286
Average	152	0.244
Excellent	98	0.158
Total	622	1.000

The contingency table for **caste** $\times$ **performance** (Table 2) reveals the strongest feature–target association in the dataset. ST-caste students are predominantly Average (93.2%), while General-caste students produce zero Average records and 26.8% Excellent — a pattern confirmed later by logistic regression feature importance.

Table 2: Contingency table: `caste` $\times$ `performance` (counts).

Caste	Avg	Good	Vg	Exc	Total
General	0	102	117	80	299
OBC	25	67	47	15	154
SC	31	20	12	3	66
ST	96	5	2	0	103
Total	152	194	178	98	622

Table 3 gives descriptive statistics for the Auction Verification dataset. The target `verification.time` is heavily right-skewed: its mean (7,337s) is $5.6\times$ the median (1,317s) and its standard deviation (10,375s) exceeds the mean, confirming that a small number of extreme auction runtimes dominate the distribution. IQR analysis places the upper fence at 26,371s; 206 instances (10.1%) exceed this threshold and are retained as legitimate observations, motivating the log transform applied to the target. Among the predictors, `process.b4.capacity` and `property.winner` are binary; the remaining features are low-cardinality integers.

Table 3: Descriptive statistics for the Auction Verification dataset ($n = 2,043$).

Feature	Mean	Std	Min	Median	Max
proc.b1.cap	1.00	0.82	0.0	1.0	2.0
proc.b2.cap	2.09	0.81	0.0	2.0	3.0
proc.b3.cap	1.88	0.32	1.0	2.0	2.0
proc.b4.cap	0.59	0.49	0.0	1.0	1.0
prop.price	71.5	8.05	59	70	90
prop.product	3.25	1.80	1.0	3.0	6.0
prop.winner	0.46	1.06	0.0	0.0	4.0
verif.time (s)	7,337	10,375	78	1,317	44,131

2.3 Pre-processing & Feature Engineering

Classification. Ordered categorical features were encoded with `OrdinalEncoder`; nominal features with `OneHotEncoder`. A `SelectKBest` filter using the χ^2 statistic was placed in the pipeline and its cutoff k was treated as a tunable hyperparameter. All numeric values were standardized with `StandardScaler`. The `time` feature was removed prior to modelling because two of its categories contained only one instance each, producing unreliable conditional distributions. The target was mapped to integers 0–3 (Average through Excellent) for compatibility with scikit-learn’s classification metrics.

Regression. All seven features were standardized with `StandardScaler`. To address the heavy right-skew in `verification.time` (mean 7,337s, max 44,130s, median 1,317s), a log transform was applied to the target via `TransformedTargetRegressor`, which automatically inverts predictions back to the original scale for evaluation.

The dataset was split 80/20 (train/test) with stratification on `property.winner` to account for its severe class imbalance.

3 Methods

3.1 Classification Task

- **kNN (Zach):** A k -Nearest Neighbors classifier preceded by χ^2 feature selection and `StandardScaler`. Tuned over $k \in \{9, 13, 15, 17, 21\}$, $n_neighbors \in \{3, 5, 7, 9, 11\}$, weight functions, and distance metrics (100 configurations, 10-fold CV). Best configuration: $n_neighbors = 11$, distance weighting, euclidean metric, $k = 9$.
- **Logistic Regression (Alan):** Multinomial Logistic Regression with ℓ_2 regularization, embedded in the same preprocessing pipeline. Tuned via `GridSearchCV` over regularization strength $C \in \{0.01, 0.1, 1, 10\}$ and feature count $k \in \{5, 7, \dots, 29\}$ (52 configurations, 10-fold CV). Best configuration: $C = 0.1$, $k = 15$.
- **Gaussian NB (Miguel):** Gaussian Naive Bayes, which models each feature as conditionally Gaussian given the class. Tuned over $\text{var_smoothing} \in \{10^{-12}, \dots, 10^{-1}\}$ (12 configurations, 10-fold CV). Best configuration: $\text{var_smoothing} = 0.1$.

3.2 Regression Task

- **kNN (Zach):** A k -Nearest Neighbors regressor with `StandardScaler` preprocessing, wrapped in `TransformedTargetRegressor` with a log transform on the target. Tuned over neighbors, weight functions, metrics, and feature count (48 configurations, 10-fold CV). Best configuration: $n_neighbors = 7$, distance weighting, manhattan metric, $k = 5$.
- **SVR (Miguel):** Support Vector Regression with an RBF kernel, also wrapped with the log-target transformer. Tuned over $C \in \{0.1, 1, 10, 100\}$, $\epsilon \in \{0.01, 0.1, 1\}$, and kernel $\in \{\text{rbf}, \text{linear}\}$ (48 configurations, 5-fold CV). Best configuration: $C = 10$, $\epsilon = 0.1$, RBF kernel.
- **Random Forest (Alan):** `RandomForestRegressor` wrapped in `TransformedTargetRegressor`. Tuned via `GridSearchCV` over $n_estimators \in \{100, 200, 500\}$, $\text{max_depth} \in \{5, 10, 20, \text{None}\}$, and $\text{min_samples_leaf} \in \{1, 5, 10\}$ (36 configurations, 10-fold CV). Best configuration: $n_estimators = 500$, $\text{max_depth} = \text{None}$, $\text{min_samples_leaf} = 1$.
- **Linear Regression:** Baseline only; not tuned, as weak cross-validation performance ($R^2 = 0.146$) reflects strong nonlinearity in the target relationship.

4 Experimental Setup and Evaluation

Both tasks used a stratified 80/20 train-test split with `random_state=42`. Model selection used 10-fold cross-validation on the training set; the test set was held out until final evaluation. `GridSearchCV` with 10-fold CV was used for hyperparameter tuning on all models per task.

Classification performance is reported as overall accuracy and macro-averaged F_1 -score. The 25% uniform-random baseline provides a lower bound for a four-class problem. Regression performance is reported as R^2 and RMSE (in seconds). Cross-validation results flag potential overfitting when the training score differs from the validation score by more than 20% relative to the validation score.

Table 4: Performance summary for both tasks. Tuned CV scores shown for all models; test scores reported for best model per task.

Task	Model	Acc / R^2	F_1 / RMSE
Class. (CV)	kNN	0.485	0.391
	Log. Reg.	0.521	0.490
	Gaussian NB	0.509	0.352
Class. (Test)	Log. Reg.	0.544	0.53
Regr. (CV)	kNN	0.952	—
	SVR	0.866	—
	Rand. For.	0.992	903 s
	Linear Reg.	0.146	—
Regr. (Test)	Rand. For.	0.991	1,078 s

5 Results & Discussion

Classification. Logistic Regression (54.4% test accuracy) substantially outperforms random chance (25%) and both alternatives. After tuning, Gaussian NB improved dramatically from 33.8% to 50.9% CV accuracy, nearly matching Logistic Regression; the default `var.smoothing` was far too small for this dataset. kNN improved from 42.8% to 48.5% CV accuracy after tuning but was flagged for overfitting at baseline; the high-dimensional one-hot encoded feature space likely degrades distance-based similarity (curse of dimensionality).

The confusion matrix (Figure 1) reveals the tuned Logistic Regression is most reliable for the majority class (Average: 83% recall) and least reliable for Excellent students (30% recall), consistent with class imbalance. The strongest predictor identified by feature importance is General caste category, followed by coaching and parental occupation features.

Limitations: the four-class target is ordinal but was treated as nominal, discarding useful ordering information. A model aware of the ordering may improve minority-class recall. **Sources of error:** class imbalance biases every model towards Average, inflating accuracy

while suppressing minority-class recall.

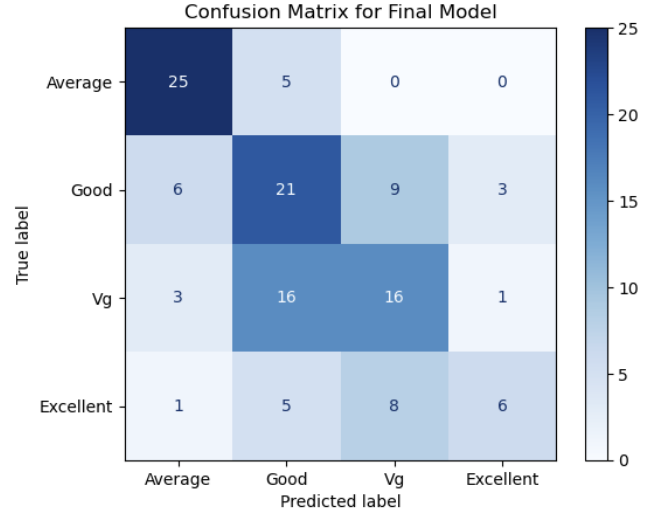


Figure 1: Confusion matrix for tuned Logistic Regression on the held-out test set. The model is most reliable for Average and least reliable for Excellent, consistent with class imbalance.

Regression. Random Forest substantially outperforms all alternatives, achieving $R^2 = 0.991$ on the test set with RMSE= 1,078s and MAE= 524s. kNN reached a strong $R^2 = 0.952$ after tuning, up from 0.853 at baseline. SVR improved most dramatically, from $R^2 = 0.493$ at baseline to 0.866 after tuning, confirming that SVR was not a weak model—it was simply under-configured with default hyperparameters. Linear Regression performed poorly ($R^2 = 0.146$), indicating strong nonlinearity in the data.

Although cross-validation flagged potential overfitting in Random Forest (train RMSE= 344s vs. CV RMSE= 903s), the small gap between CV and test performance (903s vs. 1,078s) confirms the tuned model generalizes well. Feature importance analysis identifies `property.product` (0.417) and `process.b1.capacity` (0.392) as dominant predictors, together accounting for 81% of the model’s decisions.

Limitations: the remaining gap between training RMSE (344s) and test RMSE (1,078s) indicates some overfitting persists after tuning. Random Forest predictions are less interpretable than linear models. **Sources of error:** the extreme right skew of `verification.time` (max 44,130s vs. median 1,317s) means even a well-fitted model faces large absolute errors on outlier auctions.

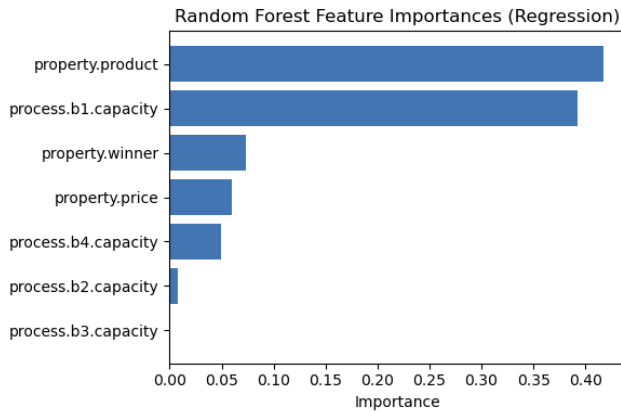


Figure 2: Random Forest feature importances for the regression task. `property.product` and `process.b1.capacity` account for 81% of feature importance.

6 Conclusion

Logistic Regression is the best-performing classifier at 54.4% accuracy, benefiting from its ability to model sparse, high-dimensional categorical data with a linear decision boundary and ℓ_2 regularization. Random Forest is the clear winner for regression ($R^2 = 0.991$), capturing the strong nonlinear interaction between auction process capacity and product type that linear methods cannot represent. A key finding across both tasks is the importance of tuning all models rather than only the best baseline: Gaussian NB and SVR each improved substantially under `GridSearchCV`, changing the narrative from “weak models” to competitive alternatives.

Contributions

- **Alan Ray:** data pre-processing, Logistic Regression, Random Forest, report writing
- **Zach Shotwell:** kNN classifier, kNN regressor
- **Miguel Moreno Coin:** Gaussian NB, SVR

References

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